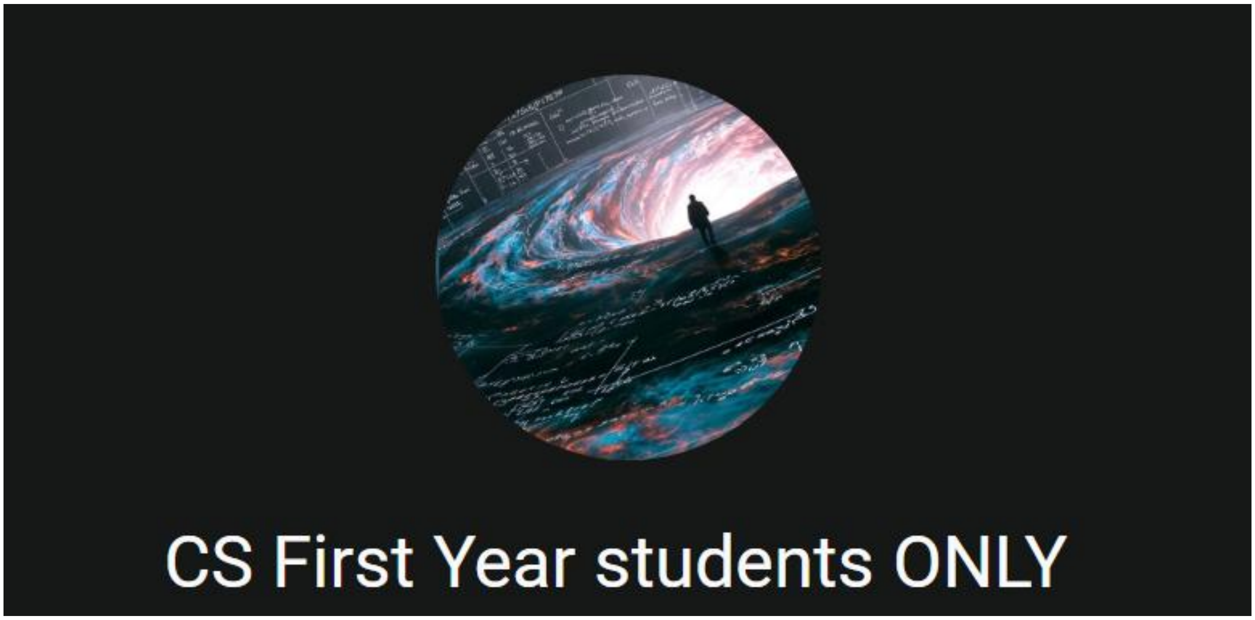
Final Exam For Second Semester (2024/2025)



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Final Exam for Second Semester (2024/2025)

Course Name: Object Oriented Programming	Course Code: CS103 Bylaw: 2024
Year: First Model: B	Program: Computer Science
Date of Exam: Wednesday 21 /05/2025	Time Allowed: 2-hours Total Marks: 60
Course Instructor: Prof. Mohamed Sayed Farag	No. of questions: (5) No. of pages: (3)

Exam instructions

- Make sure that the question form is identical to the answer sheet form
- The answer sheet must be in good condition from the moment received until it is delivered after taking the exam
- Do not bend the answer sheet or lean on it while answering

Answer the following questions:

Question One: [10 Marks], one for each. (ILOs: a1- a5, b1-b4)

Choose only one correct answer A, B, C or D for the following:

1	**What is the syntax of inheritance of class?**			
	A	class name	B	class name: access specifier
	C	class name: access specifier class name	D	None of above
2	**In OOP, data hiding is achieved using:**			
	A	Public access modifier	B	Protected access modifier
	C	Private access modifier ✓	D	Global variables
3	**Which of the following allows objects of different classes to be treated as objects of a common superclass?**			
	A	Overloading	B	Encapsulation
	C	Inheritance	D	Polymorphism
4	**If a .... is defined in the base class, it need not be necessarily redefined in the derived class**			
	A	member function	B	virtual function
	C	static function	D	real function
5	**Inheritance allows in C++ program?**			
	A	Class Re-usability	B	Creating a hierarchy of classes
	C	Extendibility	D	All above ✓
6	**Which concept allows the same function name to behave differently?**			
	A	Inheritance	B	Encapsulation
	C	Polymorphism	D	Abstraction
7	**A class can inherit properties of another class which is known as .... Inheritance.**			
	A	Single ✓	B	Multiple
	C	Multilevel	D	Hierarchical
8	**Which of the following gets called when an object is being created?**			
	A	Constructor	B	Virtual Function
	C	Destructors	D	Main
9	**Which of the following is not a principle of Object-Oriented Programming?**			
	A	Encapsulation	B	Polymorphism
	C	Inheritance	D	Compilation
10	**What is the primary purpose of being a constructor in a class?**			
	A	Destroy an object	B	Initializing an object ✓
	C	Overload an operator	D	Call a superclass

Question Two: [10 Marks], one for each (ILOs: a1- a5, b1-b5)

For each of the following statements, shade Ⓣ if the statement is True, or Ⓕ if the statement is False:

1- Virtual functions are mainly used to achieve Compile time polymorphism. T
2- The keyword used to represent a friend function is friend. T
3- Those operators which operate on two operands or data are called binary operators. T
4- Function declaration is also called a function prototype. T
5- Function call has two methods Call by value and Call by pointer.
6- A destructor is used to destroy the objects that have been created by a Constructor. T
7- When a program encounters a problem, it throws an exception. The throw keyword helps the program performs the throw. T
8- It is impossible to inherit from a template class. F
9- Template is an example of Compile time polymorphism. F
10- A C++ exception is a response to an exceptional circumstance that arises while a program is running, such as an attempt to multiply by zero. T

Question Three: [10 Marks], 2 for each (ILOs: a1- a5, b1-b5, c3)

a- **Define encapsulation and explain its advantages.**
b- **Differentiate between method overloading and method overriding.**
c- **What is the CRC Cards?**
d- **Explain the use of the virtual keyword.**
e- **What is function overloading vs. operator overloading?**

Question Four: [10 Marks], 2.5 for each (ILOs: a1, a3, b1, b3-b4 c1-c2, d1)

What is the output of the following code?

a

```
#include <iostream>
using namespace std;

class A {
public:
    void show() {
        cout << "A";
    }
};

class B : public A {
public:
    void show() {
        cout << "B";
    }
};

int main() {
    A* obj = new B();
    obj->show();
    return 0;
```

A

b

```
#include <iostream>
using namespace std;

class A {
public:
    virtual void show() {
        cout << "A";
    }
};

class B : public A {
public:
    void show() override {
        cout << "B";
    }
};

int main() {
    A* obj = new B();
    obj->show();
    return 0;
}
```

B

c	`#include<iostream>` `using namespace std;` `class demo{` `public:` `int f_num, s_num;` `sum(int a, int b){` `cout<<a+b;` `}` `};` `main(){` `demo d1;` `d1.sum(d1.f_num=10, d1.s_num=20);` `return 0;}` 30	d	`#include <iostream>` `using namespace std;` `int Add(int X, int Y, int Z)` `{` `return X + Y;` `}` `double Add(double X, double Y, double Z` `{` `return X + Y;` `}` `int main()` `{` `cout << Add(5, 6);` `cout << Add(5.5, 6.6);` `return 0;` `}` 11 , 12,1

Question Five: [20 Mark], (ILOs: a1, a3, a5, a6, b1, b3, b4, b5, c2, d1)

Design and implement the following system:
Problem: Library Management
Create a program using OOP concepts to manage a small library.
Requirements:

- **Base class:** Item with attributes: id, title, and author.
- **Derived class:** Book with additional attributes: genre and pageCount.
- **Derived class:** Magazine with attributes: issueNumber and month.
- **Functions to:**
  - Display details of items.
  - Use constructor overloading.
  - Use virtual functions.

(You must demonstrate inheritance, polymorphism, and encapsulation)

Sample Output:

Book -	ID: 1, Title: C++ Basics, Author: Bjarne, Genre: Programming, Pages: 320
Magazine -	ID: 2, Title: AI Weekly, Author: Jane Doe, Issue: 42, Month: May

End of Questions
Good Luck,
Prof. Mohamed Sayed Farag